

❖ Machine Learning Workflow Notes

(Problem Definition Feature Engineering)

These are the most important steps before training any ML model. Many beginners focus only on algorithms, but **80% of a Data Scientist's work is data preparation and understanding the problem.**

1. Problem Definition

What is Problem Definition?

Before building a model, clearly define:

What are we trying to predict?

and

Why are we predicting it?

Questions to Ask

What is the business problem?

Example:

Netflix:

Will a user like this movie?

Amazon:

Will a customer buy this product?

Bank:

Will a customer default on a loan?

What is the target variable?

Target = Output to predict

Examples:

House Price Price

Titanic Survived

Loan Approval Approved

Is it Classification or Regression?

Problem	Type
Spam Detection	Classification
House Price Prediction	Regression
Disease Detection	Classification
Sales Forecasting	Regression

Example

Problem:

Predict whether a customer will leave a company.

Target:

Churn

Type:

Classification

2. Data Collection

What is Data Collection?

Gathering relevant data for solving the problem.

Sources of Data

1. Company Databases

MySQL
PostgreSQL
Oracle

2. APIs

Examples:

- Weather API
 - Twitter API
 - Google Maps API
-

3. Public Datasets

Popular Websites:

- [Kaggle](#)
- [UCI Machine Learning Repository](#)

4. Web Scraping

```
import pandas as pd  
  
tables = pd.read_html(url)
```

Good Data Characteristics

Accurate

Relevant

Complete

Consistent

Up-to-date

Bad Data Examples

Missing Values

Wrong Age

Duplicate Rows

Outdated Information

3. Exploratory Data Analysis (EDA)

What is EDA?

EDA means understanding your dataset before building models.

Goal:

```
Find patterns  
Find anomalies  
Find relationships
```

Load Dataset

```
import pandas as pd  
  
df = pd.read_csv("data.csv")
```

First Look

```
df.head()
```

Dataset Information

```
df.info()
```

Statistical Summary

```
df.describe()
```

Missing Values

```
df.isnull().sum()
```

Correlation

```
df.corr(numeric_only=True)
```

Common EDA Visualizations

Histogram

Used for distribution.

```
import matplotlib.pyplot as plt  
  
df["Age"].hist()  
  
plt.show()
```

Box Plot

Used for outlier detection.

```
df.boxplot(column="Salary")  
  
plt.show()
```

Scatter Plot

Relationship between variables.

```
plt.scatter(  
    df["Area"],  
    df["Price"]  
)  
  
plt.show()
```

Correlation Heatmap

```
import seaborn as sns  
  
sns.heatmap(  
    df.corr(numeric_only=True),  
    annot=True  
)
```

What to Look for in EDA?

Missing Values

NULL
NaN
Empty Cells

Outliers

Very Large Salary

Negative Age

Extreme Values

Relationships

Area

Price

Positive correlation.

4. Data Cleaning

What is Data Cleaning?

Removing errors and inconsistencies from data.

Missing Values

Example:

Age

20

22

NULL

25

Fill Missing Values

```
df["Age"].fillna(  
    df["Age"].mean(),  
    inplace=True  
)
```

Remove Missing Rows

```
df.dropna(inplace=True)
```

Duplicate Records

```
df.drop_duplicates(inplace=True)
```

Wrong Data Types

Check:

```
df.dtypes
```

Convert:

```
df["Age"] = df["Age"].astype(int)
```

Outlier Treatment

IQR Method

```
Q1 = df["Salary"].quantile(0.25)
```

```
Q3 = df["Salary"].quantile(0.75)
```

```
IQR = Q3 - Q1
```

Upper Bound:

```
upper = Q3 + 1.5 * IQR
```

Lower Bound:

```
lower = Q1 - 1.5 * IQR
```

Filter:

```
df = df[
    (df["Salary"] >= lower)
    &
    (df["Salary"] <= upper)
]
```

5. Data Preprocessing

Why Preprocessing?

Machine Learning models need numerical, clean data.

Label Encoding

Convert categories into numbers.

```
from sklearn.preprocessing import LabelEncoder
```

```
le = LabelEncoder()  
  
df["Gender"] = le.fit_transform(  
    df["Gender"]  
)
```

Example:

```
Male    1  
  
Female  0
```

One Hot Encoding

Preferred for nominal categories.

```
pd.get_dummies(  
    df,  
    columns=["City"]  
)
```

Example:

```
Delhi  
Mumbai  
Pune
```

becomes

```
City_Delhi  
  
City_Mumbai
```

```
City_Pune
```

Feature Scaling

Needed for:

- KNN
 - SVM
 - Gradient Descent
 - Neural Networks
-

Standardization

```
from sklearn.preprocessing import StandardScaler  
  
scaler = StandardScaler()  
  
X_scaled = scaler.fit_transform(X)
```

6. Feature Selection

What is Feature Selection?

Selecting useful features and removing unnecessary ones.

Example

Dataset:

```
Age  
Salary  
Name  
Customer_ID
```

Useful:

```
Age  
Salary
```

Remove:

```
Name  
Customer_ID
```

Why Important?

Faster Training

Less Overfitting

Better Accuracy

Correlation Method

```
corr = df.corr(  
    numeric_only=True  
)  
  
print(corr["Price"])
```

SelectKBest

```
from sklearn.feature_selection import SelectKBest
from sklearn.feature_selection import f_regression

selector = SelectKBest(
    score_func=f_regression,
    k=5
)

X_new = selector.fit_transform(
    X,
    y
)
```

7. Feature Engineering

What is Feature Engineering?

Creating new features from existing data.

Example 1

Original Features:

```
Day
Month
Year
```

Create:

```
DayOfWeek
```

Weekend

Example 2

Original:

Date of Birth

Create:

Age

Example 3

House Dataset

Original:

Length

Width

Create:

Area

Code:

```
df["Area"] = (  
    df["Length"]  
    *  
    df["Width"]  
)
```

Why Feature Engineering Matters?

A simple model with great features often beats a complex model with poor features.

```
Good Features + Simple Model  
>  
Bad Features + Complex Model
```

Complete Workflow Example

```
import pandas as pd  
  
df = pd.read_csv("data.csv")  
  
# Cleaning  
df.drop_duplicates(inplace=True)  
  
df.fillna(df.mean(numeric_only=True),  
          inplace=True)  
  
# Encoding  
df = pd.get_dummies(  
    df,  
    drop_first=True  
)  
  
# Feature Engineering  
df["Area"] = (  
    df["Length"]  
    *  
    df["Width"]  
)  
  
# Features  
X = df.drop("Price", axis=1)
```



```
# Target  
y = df["Price"]
```

Interview Revision (1-Minute)

Problem Definition

- Define objective
- Identify target variable
- Classification or Regression

Data Collection

- Database
- APIs
- Kaggle
- Web Scraping

EDA

- Understand data
- Find patterns
- Detect issues

Data Cleaning

- Missing values
- Duplicates
- Outliers

Preprocessing

- Encoding
- Scaling

Feature Selection

- Keep useful features
- Remove irrelevant features

Feature Engineering

- Create new meaningful features
- Often improves performance more than changing algorithms

This workflow is the foundation of every ML project, whether you're building a Titanic classifier, house-price predictor, recommendation system, or advanced AI model.